

Deep Learning for Natural Language Processing

Assignment- 7

TYPE OF QUESTION: MCQ

Number of questions: 10
each]

Total marks: 10 [1 mark

Question 1: A TF-IDF retriever uses the simplified formulation from the lecture, where the term frequency is squashed as $tf(t, d) = \log_{10}(\text{count}(t, d)) + 1$ and the inverse document frequency is $idf(t) = \log_{10}(N / df(t))$.

A collection contains $N = 10,000$ documents. The term "neural" occurs in $df = 100$ documents, and occurs 100 times inside document d . What is the tf-idf weight of "neural" in d ?

- A. 3.0
- B. 4.0
- C. 6.0
- D. 2.0

Answer: C

$$tf = \log_{10}(100) + 1 = 2 + 1 = 3.$$

$$idf = \log_{10}(10000 / 100) = \log_{10}(100) = 2.$$

$$tf-idf = 3 \times 2 = 6$$

Question 2: SQuAD is evaluated using exact match (0 or 1) and F1 (partial credit), where the score against each of the 3 gold answers is computed and the maximum is taken. Articles (a, an, the) and punctuation are stripped before matching.

Gold answers: {"the Bengal tiger", "a Bengal tiger", "a Bengal tiger in India"}

Prediction: "the Bengal tiger habitat"

What are the Exact Match and F1 scores for this example?

- A. EM = 0, F1 = 0.57
- B. EM = 0, F1 = 0.80
- C. EM = 1, F1 = 1.00
- D. EM = 0, F1 = 0.67

Answer: B

After normalization (articles removed), the prediction is {bengal, tiger, habitat} (3 tokens).

The gold answers become {bengal, tiger} (2 tokens), {bengal, tiger} (2 tokens) and {bengal, tiger, in, india} (4 tokens).

No gold answer is identical to the prediction, so EM = 0. Using $F1 = 2 \cdot \text{overlap} / (|\text{pred}| + |\text{gold}|)$:

vs gold 1: overlap = 2 $\rightarrow F1 = 2(2)/(3+2) = 0.80$

vs gold 2: overlap = 2 $\rightarrow F1 = 0.80$

vs gold 3: overlap = 2 $\rightarrow F1 = 2(2)/(3+4) = 0.571$

Taking the maximum over the three gold answers gives $F1 = 0.80$.

Question 3: ColBERT scores a query–document pair by late interaction:

$$\text{score}(q, d) = \sum_{i=1}^N \max_{j=1}^m \mathbf{E}_{q_i} \cdot \mathbf{E}_{d_j}$$

For a query of $N = 32$ tokens and a passage of $m = 180$ tokens, how many token-level dot products must be computed, and how many values are finally summed to produce $\text{score}(q, d)$?

- A. 5760 dot products; 32 values summed
- B. 5760 dot products; 180 values summed
- C. 212 dot products; 32 values summed
- D. 32 dot products; 180 values summed

Answer: A

Late interaction computes the full $N \times m$ similarity matrix: $32 \times 180 = \mathbf{5760}$ dot products.

A max is then taken along the document axis **for each query token**, producing $N = 32$ MaxSim values, which are summed to give $\text{score}(q, d)$.

Note that the document token embeddings can be pre-computed and indexed offline (unlike a cross-encoder), which is what makes this tractable.

Question 4: BM25 scores a document as

$$\text{score}(q, d) = \sum_{t \in q} \overbrace{\log \left(\frac{N}{df_t} \right)}^{\text{IDF}} \overbrace{\frac{tf_{t,d}}{k \left(1 - b + b \left(\frac{|d|}{|d_{\text{avg}}|} \right) \right) + tf_{t,d}}}^{\text{weighted tf}}$$

with the lecture's default parameters $k = 1.2$, $b = 0.75$. A one-term query has $N/df_t = 100$, the raw term frequency in the document is $tf_{t,d} = 5$, and the document is exactly twice the average document length.

What is the BM25 score (to 2 decimals)?

- A. 1.41
- B. 2.00
- C. 1.19
- D. 0.70

Answer: A

IDF part: $\log_{10}(N/df_t) = \log_{10} 100 = 2$.

Length normalization: $|d|/|d_{\text{avg}}| = 2$, so

$k(1 - b + b \cdot 2) = 1.2 \times (1 - 0.75 + 0.75 \times 2) = 1.2 \times (0.25 + 1.5) = 1.2 \times 1.75 = 2.1$

Weighted tf:

$tf / (2.1 + tf) = 5 / (2.1 + 5) = 5 / 7.1 = 0.7042$

score = $2 \times 0.7042 = 1.4085 \approx 1.41$.

Question 5: A reference has 3 tokens with idf weights [1.0, 2.0, 3.0] and greedy-matched maximum cosine similarities [0.9, 0.6, 0.4].

What are the idf-weighted and the plain (uniform) BERTScore recall values?

- A. weighted 0.55, uniform 0.63
- B. weighted 0.63, uniform 0.55
- C. weighted 0.55, uniform 0.55
- D. weighted 0.73, uniform 0.63

Answer: A

IDF-weighted BERTScore recall:

$R = (1.0 \times 0.9 + 2.0 \times 0.6 + 3.0 \times 0.4) / (1.0 + 2.0 + 3.0)$

$$= (0.9 + 1.2 + 1.2) / 6 = 3.3 / 6 = 0.55$$

Uniform (plain BERTScore recall):

$$R = (0.9 + 0.6 + 0.4) / 3 = 1.9 / 3 = 0.633 \approx 0.63$$

Question 6: A DPR bi-encoder is trained with a batch of $B = 8$ questions. Each question is paired with one gold positive passage and one BM25 hard negative. All passages in the batch are shared, so every question uses every other passage in the batch as a negative.

How many negative passages does each question have, and how many question–passage dot products must be computed for the batch?

- A. 7 negatives, 64 dot products
- B. 8 negatives, 64 dot products
- C. 15 negatives, 128 dot products
- D. 14 negatives, 128 dot products

Answer: C

The batch contains $B \times 2 = 8 \times 2 = 16$ passages (8 positives + 8 hard negatives). For a given question, exactly one of those 16 is its own positive, so the remaining $16 - 1 = 15$ passages act as negatives.

The similarity matrix needed is every question against every passage: $8 \times 16 = 128$ dot products.

Question 7: In an open-domain QA pipeline, the retriever's recall@20 is 0.80. The reader produces the correct answer 75% of the time when the gold passage is retrieved, and 5% of the time when it is not (lucky guesses from parametric memory).

What is the expected end-to-end Exact Match of the pipeline?

- A. 60.0%
- B. 61.0%
- C. 75.0%
- D. 80.0%

Answer: B

Law of total probability over the retrieval outcome:

$$\begin{aligned}
EM &= P(\text{retrieved}) \times P(\text{correct} \mid \text{retrieved}) + P(\text{not retrieved}) \times P(\text{correct} \mid \text{not retrieved}) \\
&= 0.80 \times 0.75 + 0.20 \times 0.05 \\
&= 0.60 + 0.01 \\
&= 0.61 \rightarrow 61.0\%
\end{aligned}$$

Question 8: A TF-IDF retriever uses the definitions: $\text{tf}(t,d) = \log_{10}(1 + \text{count}(t,d))$, $\text{idf}(t) = \log_{10}(N / \text{df}(t))$, and scores by cosine similarity of the tf-idf vectors.

The collection has $N = 1000$ documents. Query $q = \text{"neural network"}$ (each term occurs once in the query).

Term	df	Count in document d
neural	100	9
network	10	99

Only these two terms are non-zero in both vectors. What is $\text{score}(q, d) = \cos(q, d)$?

- A. 0.976
- B. 0.707
- C. 1.000
- D. 0.812

Answer: A

Document tf-idf weights:

neural: $\text{tf} = \log_{10}(1+9) = \log_{10} 10 = 1$; $\text{idf} = \log_{10}(1000/100) = 1 \rightarrow \text{tf-idf} = 1 \times 1 = 1.0$

network: $\text{tf} = \log_{10}(1+99) = \log_{10} 100 = 2$; $\text{idf} = \log_{10}(1000/10) = 2 \rightarrow \text{tf-idf} = 2 \times 2 = 4.0$

Query tf-idf weights (count = 1, so $\text{tf} = \log_{10} 2 = 0.301$ for both):

neural: $0.301 \times 1 = 0.301$

network: $0.301 \times 2 = 0.602$

So $q \propto (1, 2)$ and $d \propto (1, 4)$. Cosine is scale-invariant, so:

$$\cos = (1 \cdot 1 + 2 \cdot 4) / (\sqrt{1^2 + 2^2} \cdot \sqrt{1^2 + 4^2}) = 9 / (\sqrt{5} \cdot \sqrt{17}) = 9 / \sqrt{85} = 9/9.2195 = 0.976$$

Question 9: A 3-token hypothesis has token log-probabilities $[-0.5, -1.0, -2.0]$. Compute BARTScore under (i) uniform weights and (ii) idf weights $[0.5, 1.0, 1.5]$ normalized to sum to 1.

- A. (i) -1.17 , (ii) -1.42
- B. (i) -3.50 , (ii) -4.25
- C. (i) -1.42 , (ii) -1.17
- D. (i) -1.17 , (ii) -4.25

Answer: A

(i) Uniform weights $\omega_t = 1/3$:

$$\text{BARTScore} = (1/3)(-0.5 - 1.0 - 2.0) = -3.5/3 = -1.1667 \approx -1.17$$

(ii) IDF weights $[0.5, 1.0, 1.5]$, normalized (sum = 3) $\rightarrow \omega = [1/6, 1/3, 1/2]$:

$$\begin{aligned}\text{BARTScore} &= (1/6)(-0.5) + (1/3)(-1.0) + (1/2)(-2.0) \\ &= -0.0833 - 0.3333 - 1.0000 \\ &= -1.4167 \approx -1.42\end{aligned}$$

Question 10: During the decoding phase of a Dialogue System, temperature scaling is applied to control the diversity of the generated response. The unnormalized logits for the vocabulary $V =$

$\{w_1, w_2, w_3\}$ are $[\ln(4), \ln(2), 0]$. If a temperature of $T = 0.5$ is applied before the softmax function, what is the probability of selecting the word w_1 ?

- A. $4/7$
- B. $16/21$
- C. $2/3$
- D. $4/5$

Answer: B

Temperature scaling divides the logits by T before exponentiation. The scaled logits are $[\ln(4)/0.5, \ln(2)/0.5, 0/0.5] = [2 \ln(4), 2 \ln(2), 0] = [\ln(16), \ln(4), 0]$.

Taking the exponential yields $[16, 4, 1]$.

The softmax probability for w_1 is $16/(16 + 4 + 1) = 16/21$.

